

IBRAHIM REFAAT IBRAHIM

Front-End Developer

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Portfolio | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Results-driven Front-End Developer with 3+ years of experience delivering production-grade web applications using React.js, Next.js, TypeScript, Redux Toolkit, and Firebase. Shipped two fully deployed, real-user systems for a national robotics championship — a real-time scoring platform and a QR-based attendance system — both migrated to React with advanced state management, custom hooks, and component architecture. Experienced engineering mentor and team leader with a track record of building scalable UIs and driving team performance.

TECHNICAL SKILLS

- Frameworks & Libraries: React.js | Next.js 14 (App Router) | Redux Toolkit | React Query | React Router v6 | Framer Motion | Tailwind CSS | Bootstrap
- Languages: TypeScript | JavaScript (ES6+) | HTML5 | CSS3 | Python | Java | C++ | C#
- Backend & APIs: Firebase (Firestore, Auth, Realtime DB, Hosting) | REST APIs | MySQL | EmailJS
- Tooling & Workflow: Vite | Git | GitHub | CI/CD | Jest | React Testing Library | Zod | React Hook Form
- Practices: Component Architecture | Custom Hooks | Performance Optimization | Responsive Design | OOP | Agile/Scrum

PROFESSIONAL EXPERIENCE

Programming Competitions Manager | Global Robotics Challenge (GRC) — Egypt2025 • Cairo, Egypt

- Architected and deployed a React + TypeScript real-time scoring system using Redux Toolkit, React Router v6, and Firebase Firestore — supporting 13 competitions across 38 categories with live score updates.
- Implemented role-based access control (Admin, Judge, Public Viewer) via React Context + Firebase Auth, with mission-based criteria scoring, audit logging, and protected route guards.
- Built a React QR attendance system using react-qr-code and html5-qrcode for 170+ teams — with automated QR generation, EmailJS delivery, and real-time Firestore check-in tracking, cutting manual check-in time by ~80%.
- Leveraged React Query for server-state caching and optimistic UI updates; deployed both apps on Firebase Hosting with CI/CD pipeline and Vite build tooling.
- Managed full event programming track: competition structure, task design, and all technical logistics for the national robotics and coding segments.

Team Leader & Technical Supervisor — Programming | Discovery Academy for Science & Technology2024 – Present • Cairo, Egypt

- Promoted within one year from Instructor to Team Leader, overseeing programming education operations and a team of instructors across the branch.
- Standardized React.js and JavaScript curriculum roadmaps, improving student project completion rates by ~40%.
- Mentored junior instructors through code reviews and project-based pedagogy, increasing student retention and satisfaction across all cohorts.
- Delivered programming courses (HTML, CSS, JavaScript, Python, OOP) to students aged 12+, consistently receiving high satisfaction ratings.

Programming Instructor (Part-time) | Zoser InstituteJul 2023 – Feb 2024 • Cairo, Egypt

- Designed and delivered hands-on web development courses using a project-first approach, helping students ship real websites within the course timeline.
- Produced structured lesson plans that reduced curriculum preparation time by ~30% for future cohorts.

Front-End Developer Intern | Sphinx Company *Apr 2023 – Oct 2023 • Cairo, Egypt*

- Built 3+ interactive React.js web applications from design spec to deployment within an Agile team, participating in sprint planning and code reviews.
 - Optimized component re-renders and implemented lazy loading, improving estimated page load speed by ~25% on targeted modules.
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PROJECTS

GRC Real-Time Scoring System | React · TypeScript · Redux Toolkit · Firebase | Production

- Rebuilt from Vanilla JS to React + TypeScript: Redux Toolkit for global score state, React Router v6 for protected role-based routes (Admin / Judge / Public), React Query for real-time Firestore sync with optimistic updates.
- Supports 13 competitions, 38 categories — deployed on Firebase Hosting with Vite build pipeline; served live judges and spectators throughout the national event.

GRC QR Attendance System | React · TypeScript · Firebase · react-qrcode | Production

- Migrated to React + TypeScript: react-qrcode for generation, html5-qrcode for camera scanning, React Hook Form + Zod for validated team registration, Framer Motion for animated scan feedback.
- Managed real-time check-in for 170+ teams via Firebase Realtime DB with EmailJS-triggered QR delivery; reduced event check-in time by ~80%.

Personal Portfolio | React · TypeScript · Tailwind CSS · Framer Motion | Vercel

- Built with React + Vite + TypeScript featuring Framer Motion animations, Tailwind CSS styling, and Vercel deployment — achieving high Lighthouse performance and SEO scores.

Sign Language Detection System | Python · TensorFlow · Keras · OpenCV | Graduation Project

- Engineered a real-time sign language-to-speech recognition model; applied data augmentation and hyperparameter tuning to boost accuracy by ~20% over the baseline prototype.
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EDUCATION

B.Sc. in Computer Science | Benha University *2018 – 2022 • Cairo, Egypt*

CERTIFICATIONS & COURSES

- Pro Frontend Engineer: React.js + TypeScript + Redux Toolkit (Udemy, Muhammad Naga) — 2025 | Jest, RTL, CI/CD, scalable architecture & testing
 - Web Development Diploma — Russian Culture Center, Cairo | 2023 | Full-stack fundamentals, responsive design, UX principles
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LANGUAGES

- Arabic — Native | English — Professional Working Proficiency